

EXERCISES



PART A – COMPETENCY FOCUSED QUESTIONS

I. Multiple Choice Questions

Tick (✓) the correct answer:

1.

pow()

ceil()

rint()

cos()

abs()

What does the above image illustrate in Java?

- (a) Mathematical methods (b) Arithmetical methods
(c) Arithmetical operators (d) Logical methods
2. Which of the following does not find the square of a number?
(a) Math.pow(a,2) (b) a*a
(c) Math.sqrt(a,2) (d) All of these
3. What type of value is returned by Math.sqrt()?
(a) int (b) float (c) double (d) All of these
4. Which of the following syntax is the correct one to find the square root of a number?
(a) sqrt(a) (b) Math.sqrt(a) (c) Squareroot(a) (d) None of these
5. Name the class that is used for different Mathematical functions. [ICSE-2007]
(a) Java.Math (b) Java.Power (c) Java.Sqrt (d) None of these
6. Give the output of Math.abs(x); when x = -9.99 [ICSE-2008]
(a) -9.99 (b) 9.99 (c) 0.99 (d) None of these
7. Give the output of Math.sqrt(x); when x = 9.0 [ICSE-2008]
(a) 3 (b) 3.0 (c) 3.00 (d) All of these
8. Read the following text and choose the correct answer:
Math.round() method returns the value of a number rounded to its nearest integer. If the fractional part is less than 0.5 then it returns the same integer value, otherwise it returns the next higher integer.
What type of data does Math.round() return?
(a) int (b) long (c) float (d) double

II. Predict the output:

1. System.out.println(Math.sqrt(10.24)); :
2. System.out.println(Math rint(-99.4)); :
3. System.out.println(Math.cbrt(42.875)); :
4. System.out.println(Math.min(-25.5, -12.5)); :
5. System.out.println(Math.ceil(-0.95)); :
6. System.out.println(Math.round(-18.51)); :
7. System.out.println(Math.max(-77.66, -87.45)); :
8. System.out.println(Math.floor(-0.88)); :

9. `System.out.println(Math rint(98.5));` :
10. `System.out.println(Math.ceil(65.5));` :

III. Assertion and Reasoning based question:

Assertion (A): `Math.sqrt( )` cannot be applied to find the square root of a negative number.

Reason (R): This function will result in NaN for a negative number.

Based on the above discussion, choose an appropriate statement from the options given below:

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not a correct explanation of Assertion(A).
- (c) Assertion (A) is true and Reason (R) is false.
- (d) Assertion (A) is false and Reason (R) is true.

IV. Application based questions:

In the following mentioned below, some of the Java statements (using different mathematical functions) are given. Punit was unable to predict the outputs.

- (a) `System.out.println(Math.max(8, Math.abs(-12)));`
- (b) `System.out.println(Math.pow(2, Math.sqrt(9)));`
- (c) `System.out.println(Math.ceil(Math.min(13.67, 8.42)));`
- (d) `System.out.println(Math.rint(6.5));`

Help your friend to predict the output in each of the above statements.

V. Write down the syntax for the following functions:

1. To find the smaller number between the two numbers p and q.

.....

2. To find the absolute value of a number m.

.....

3. To find the exponent of a number k.

.....

4. To find the square root of a number d.

.....

5. To find the rounded-off value of a number b.

.....

VI. Predict the return data type of the following functions:

- 1. `Math.sqrt( )`; :
- 2. `Math.rint( )`; :
- 3. `Math.ceil( )`; :
- 4. `Math.round( )`; :
- 5. `Math.floor( )`; :
- 6. `Math.log( )`; :

VII. Unsolved Java Programs:

1. Write a program in Java to input three numbers and display the greatest and the smallest of the two numbers.

[Hint: Use Math.min() and Math.max() function]

Sample Input: 87, 65, 34

Sample Output: Greatest number 87
Smallest number 34

2. Write a program in Java to calculate and display the hypotenuse of a right-angled triangle by taking perpendicular and base as inputs.

[Hint: $h = \sqrt{p^2 + b^2}$]

3. Write a program to input a number. Evaluate and display the results based on the number entered by the user:

(a) Natural logarithm of the number

(b) Absolute value of the number

(c) Square root of the number

(d) Cube of the number

(e) Random numbers between 0 (zero) and 1 (one).

[ICSE-2006]

4. In an examination, you have appeared for three subjects i.e. Physics, Chemistry and Biology. Write a program in Java to calculate the average marks obtained and finally display the marks in rounded-off form.

Take Physics, Chemistry and Biology marks as inputs.

5. You want to calculate the radius of a circle by using the formula:

Area = $\frac{22}{7}r^2$; where r = radius of the circle

Hence, the radius can be calculated as:

$$r = \sqrt{\frac{7 \times \text{area}}{22}}$$

Write a program in Java to calculate and display the radius of a circle by taking the area as input.

PART B – SUBJECTIVE QUESTIONS

I. Explain the following functions:

1. Math.random()

2. Math.max()

3. Math.cbrt()

4. Math.abs()

5. Math.log()

II. Distinguish between them with suitable examples:

1. Math.ceil() and Math.floor()

2. Math rint() and Math.round()